

Feature Selection in Machine Learning: Methods and Comparison

Amandeep Kaur
Chitkara University Institute of
Engineering and Technology,
Chitkara University, Punjab,
India

Kalpna Guleria*
Chitkara University Institute of
Engineering and Technology,
Chitkara University, Punjab,
India
kalpna@chitkara.edu.in

Naresh Kumar Trivedi
Chitkara University Institute of
Engineering and Technology,
Chitkara University, Punjab,
India

Abstract— Nowadays, a huge amount of data is generated every day in continuous manner in every hour and if the data is not utilized in the right or meaningful manner then this is just like garbage. Therefore, the meaningful information from the data can be represented through the feature selection. Feature selection refers to various methods which selects the most appropriate from the data according to the problem. This paper provides an insight into various feature selection methods which include filter, wrapper, embedded, hybrid and a detailed explanation of various techniques utilized by these methods. Further, a comparison among these feature selection methods have also been made.

Keywords—Feature Selection, Filter Method, Wrapper Method, Embedded Method, Hybrid feature selection,

I. INTRODUCTION

Rapid growth of Technology has increased the usage of the computers and internet which continuously produced the data in huge amounts in form of the video's, text documents, images, and voice. This data is obtained from cloud, IOT, social relations and different resources [1-3]. This vast amount of the data has an important characteristic of high dimensions, which is really a big challenge for existing machine learning approaches for data analysis and for decision making. Further, the presence of noise, irrelevant and redundant data makes the algorithms very slow and even degrades the performance of the learning task [4]. Therefore, it is very necessary to find out the useful or important features from this huge amount of the data. Hence, feature selection is a very effective method to solve this problem, by selecting the small subset of the features from this vast amount of data according to the problem. Feature selection that are useful or important to solve the problem of particular domain and removing the irrelevant features from the given dataset [1][6-7]. Basically, machine learning is classified into three families, first one is supervised method, second one is

semi-supervised method and third one is unsupervised method. Feature selection technique can pre-process the given dataset, which helps to improve the performance of the algorithms and simplify the learning and results by finding the subset of the given feature. In supervised learning the subset of the feature are selected on the basis of relevance and the important measuring criteria of the features is by the use of labelled data to train the feature selection model [8-13]. It is apparent that if in a given dataset the classes are not labelled the unsupervised feature selection is suitable. In the semi-supervised category the some labelled data integrated into unlabelled data as supplementary information to improve the performance [4] [14-16].

According to searching strategies and relationship with learning methods feature selection can be classified into four categories: filter, wrapper, embedded and hybrid [1][4]. In the filter method the characteristics of the training data are used to select the most discriminative features. In this method the features are ranked with some criteria and after that only high ranked features are selected [4][17]. A greedy search strategy is applied by wrapper method in order to retrieve the most appropriate set of features to obtain the best results [18-19]. Embedded models perform feature selection in the process of model construction. Embedded method is a built-in method of feature selection. It is embedded with a learning algorithm and overlooks the repetitive execution of algorithms and monitors the feature evaluation [20]. The fig. 1 describes the taxonomy of various feature selection methods. The hybrid method utilizes the combination of techniques used in filter and wrapper or the combination of techniques used in embedded and wrapper.

The paper is organized as follows: Section 1 covers the introduction part, section II elaborates the related work and section III explains about various methods of the feature selection and their comparison. Section IV concludes the paper.

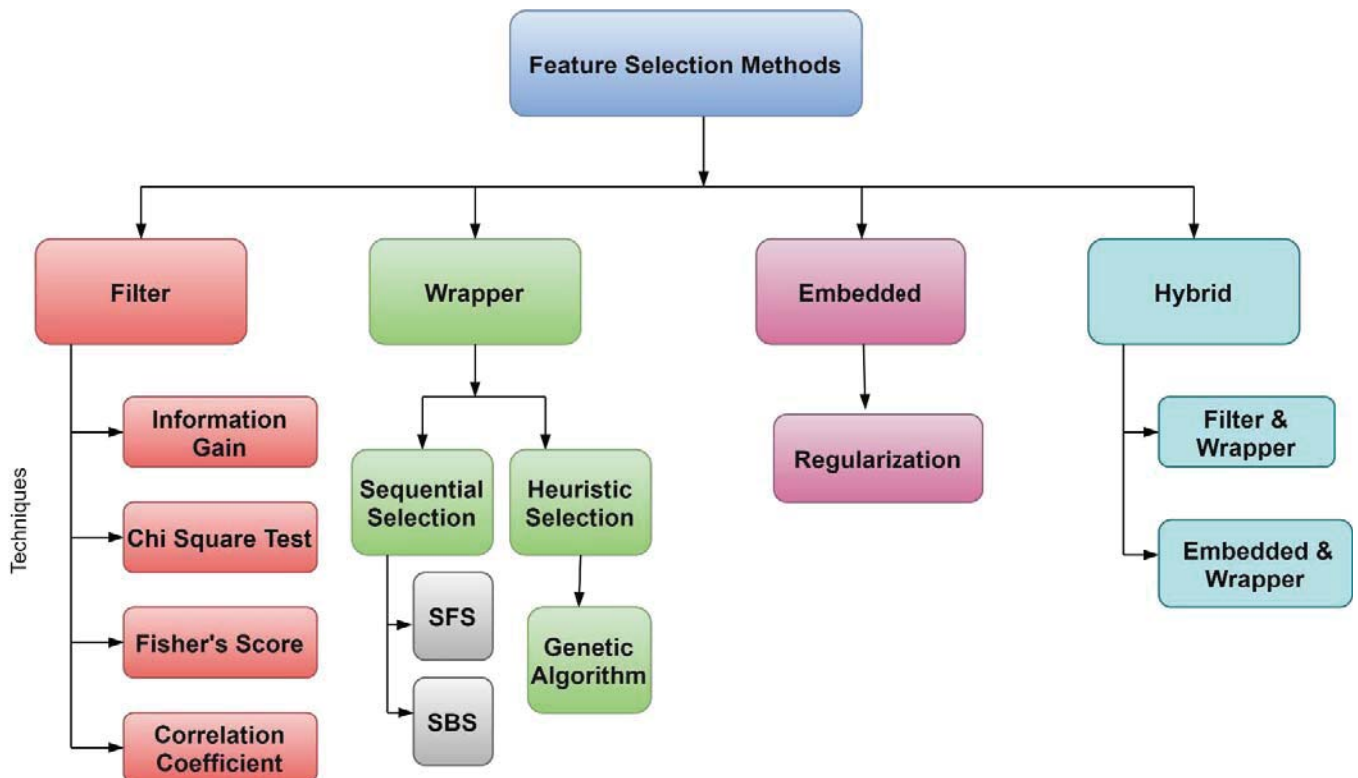


Fig. 1 Taxonomy of Feature Selection in Machine Learning

II. Related Work

In 2013 Taghi M. Khoshgoftaar et. al [21] presented a survey on the stability analysis of feature subset selection techniques. The authors focused on stability and described the importance, issues and different stability measures of evaluating the feature subset selection based on filter, embedded and wrapper techniques.

In 2014 R.K. Bania [22] presented the study on the feature selection which is basically focused on the supervised techniques of feature selection ;wrapper, embedded, hybrid, filter and searching approaches (informed search, randomized search) with the outline of the comparative study. The main objective of this survey was to represent the comprehensive and structured study of the existing methods and to describe the pros and cons of the feature selection methods. Further, the real world applications and tools of the feature selection were also discussed.

In 2015 the Bing Xue et al. [5] provided a survey of evolutionary computation for feature selection. In their survey they identify the contribution of the different algorithms, representation, search mechanism, performance measure and applications also. De-Rong Liu et al. [23] presented a study of feature selection and feature learning for high-dimensional batch reinforcement learning. Authors, provided a comprehensive survey on the progress of automatic feature selection and unsupervised learning of features for high dimensional batch reinforcement Learning.

Shahana A H et al. [24] discussed the feature subset selection for high dimensional data in the year 2016. They describe a new clustering based feature subset selection algorithm. These algorithms remove the irrelevant details from the data by minimum spanning tree construction using the tree partitioning for subsets of the subset. The overview of the different methods like wrapper method, embedded method, filter method and feature selection using FAST algorithm has been provided which is based on the feature subset selection Method and improves classifier performance. It has been proved that FAST algorithms are better than the wrapper and filter method [24][25]. Feature selection was used to implement different fields like Josef Suto et al. [26]. Further, the authors have discussed the work on human activity recognition problem in which they have used a simple naïve Bayesian wrapper method and compared it with already used methods or with algorithms on HAR (human activity recognition) machine.

In 2017, JUNDONG LI et al. [27] discussed the feature selection from the data perspective. Authors present the structured overview on the current advances in the feature selection; they review the algorithms of the feature selection for the conventional, structured, heterogeneous and streaming data. In their work they categorized on the basis of similarities, information theoretical basis, sparse learning basis and statistical based methods. A. Sheik Abdullah et al.[28] discussed

evolutionary techniques in feature selection and they have presented many feature selection algorithms using evolutionary techniques.

In 2018 Jie Cai et. al. [1] have provided a survey on feature selection in machine learning. In this survey, authors have elaborated many frequently used evaluation measures for feature selection. They have discussed classification and clustering under supervised and unsupervised feature selection methods which are applied most of the time in the machine learning problems [1][29].

In 2019 N. Krishnaveni and V. Radha [30] elaborated the algorithms of feature selection for data mining classification. Authors described the process of feature selection and the wrapper, filter and hybrid algorithms. Further, the analysis of existing algorithms of the feature selection and challenges have been discussed.

III. Methods of feature selection and Comparison

The right selection of the features is important to improve the accuracy and efficiency. Some methods of the feature selection are mentioned below:

- A. Filter Method
- B. Wrapper Method
- C. Embedded Method
- D. Hybrid Method

A. Filter Method

The filter method of the feature selection is a computationally efficient method. It is faster and less expensive when we are dealing with high dimensional data. This method can skip variables that are not beneficial but can be useful when it is combined with other [31]. This method uses ranking technique for the selection of the feature variable. The ranking technique is easy and has good results for real-world application. A suitable ranking principle is used to score the variable and threshold used to remove the variables below the threshold [32][19]. In the filter method the selection of the features is classifier independent. Filter method computes the feature selection as the pre-processing step without using any induction algorithm that is why this method is called an independent method. This method is faster than the wrapper method as it is independent from the induction algorithm [33]. Filter method uses the following techniques:

- Information Gain: Information gain is a technique in which the entropy of the dataset is reduced with the help of transformation of the dataset. Information gain is basically used to create the decision tree from the dataset which is used for the training through evaluation of the information gain of each variable. Thereafter, the variable which reduces the entropy and increases the information gain is selected. Decision tree divided the dataset into the different clusters to for the classification [34].

- Chi Square Test: The chi square test is applied to the group of the categorical features to evaluate the correlation in the dataset. In this test, we calculate the chi square between each feature and the target attribute. Further, it selects one of the best chi square from them which scores good chi square. In order to apply the chi square to evaluate the relation between the features of the dataset and the target variable there are some steps to follow [35]. The variable must be a categorical, independent and sampled. The chi square equation is as follows.

$$X_c^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

In this equation the E is expected value and O is observed value and c is degrees of freedom.

- Fisher's score: Fisher's score is also a very popular technique in the feature selection. This technique is used in the classification feature selection methods by returning the ranks of the variable based on the score received from the fisher's algorithm and it returns the rank in descending order. The feature variables will be selected on the basis of the score and rank.
- Correlation Coefficient: Through this technique we find the correlation between the two or more variables and predict. In this way, we are able to find the variables which are best suited and will be highly correlated with the target variable [36]. Suppose we have two variables A and B and the target variable is T. If the absence and presence of the A and B will be affected by the T which is the target variable that means these A and B variables are highly correlated with the target T variable. The below mentioned equation provides correlation coefficient:

$$r_{xy} = \frac{\sum_{i=1}^n (x^i - \bar{X})}{\sqrt{\sum_{i=1}^n (x^i - \bar{X})^2 \sum_{i=1}^n (y^i - \bar{Y})^2}}$$

It is used to summarize the strength of linear relationship between two data variables. 1 denotes the positive correlation and -1 denotes the negative and 0 denotes no correlation between two variables.

B. Wrapper Method

A greedy search strategy is applied by wrapper method in order to retrieve the most appropriate set of features to obtain the best results [37]. Wrapper method basically used two types of the algorithm first one is Sequential selection algorithm and second is Heuristic search algorithm. The sequential search selection algorithm will start with the empty set and the features will be removed and added until the objective function is obtained and

the heuristic search algorithms basically works on evaluation of the data set to gain the objective function..

Sequential Selection Algorithm

SFS (Sequential Forward Selection): Sequential algorithms work in an iterative manner therefore this algorithm is called a sequential algorithm. This algorithm starts with an empty set and adds the features as per the requirement. This algorithm initializes the variables which are performing well against the target variable then the next variables will be added in the subset and are evaluated. The variable will be added permanently if it provides the more accurate results than the previous one. This process is repeated till the necessary features are added.

- SBS (Sequential Backward Selection): The creation of the sequential backward selection algorithm is similar to the SFS but SBS starts with the complete variables instead of an empty set and will remove those variables which have very less impact on the performance of the predictor.

Heuristic search Algorithm

- GA (Genetic Algorithm): The Genetic algorithm is very popular in feature selection. This is basically an approximation technique which is based on the heuristic search method [38]. Genetic algorithms work on the phenomenon of the abstraction of the biological evaluation to solve the complex problem in the feature selection. The genetic algorithms operate on the population of the individuals to produce better approximation. Steps of the genetic algorithm are: Generate initial population, calculate fitness function, selection, crossover and mutation.

C. Embedded Method

The main motive of the embedded method is to reduce the computation time taken by the wrapper method in reclassifying the different subset [39]. The embedded method selects the feature in between the time when the model is under the training. The main advantage of the embedded method is that the algorithm in the embedded method can perform the selection of the variables, selection of the feature and classification in the one time. Embedded method is faster and more accurate than the filter method. In this method the

machine learning model is generated first and it fetches the features after the measure which are important for the prediction. Further, it removes the features which are less important for the target or for the prediction.

Regularization: It includes LASSO, RIDGE and Elastic NET. The regularization mainly consists of adding the different parameters of the machine learning to overcome the under fitting or high bias.

D. Hybrid Method

In the hybrid approach more than one methods or approaches are combined to achieve the good feature subset [40]. In other words we can say that the elimination or reduction of the features will start with the filter method and in the next step to selection of the feature which are important for the prediction is through the wrapper method. This approach is a simple and high level. The hybrid method is more accurate and performs well than the wrapper method and computational time taken by the hybrid method is better than the wrapper method. The main advantage of the hybrid method is that it takes the advantages from the other feature selection methods which gives the better results. The process to combine the methods is as follows.

This is very important to choose which method is used in combination to achieve your target. Suppose, filter and wrapper method are selected to combine as a hybrid approach ; then the techniques of the filter method will be utilized for ranking like chi score, information gain to order the features without involving any learning algorithm. Thereafter, selection of the best feature from the generated ordered list will be done through the wrapper method like SBS or SFS. So, in this way the combined approaches can be utilized as hybrid methods. Similarly, the combination of the Embedded and Wrapper approach can be used together as a hybrid. Table 1 shows the summary and comparison of various feature selection methods along with advantages and research gaps of the same.

Table 1. The Summary and comparison of Feature Selection Methods

Method	Techniques	Summary	Advantages	Research Gaps
Filter [34][35]	Information Gain	It evaluates the information gain of each variable. It reduces the entropy of the dataset and divide the dataset into clusters for classification. Equation to calculate entropy of dataset is : $E = - \sum_i^c p_i \log_2 p_i$	It is fast and simple method and it has no dependency the classifier.	The drawback of this method is that it considers do not considers the dependencies among the features as individual features are considered independently
	Chi Square	Evaluate the relationship between the variables of the data set and target variable as per the best chi-square score. Formula of chi Square. $\chi_c^2 = \sum \frac{(O_i - E_i)^2}{E_i}$		
	Fisher's Score	Variables are selected as per the ranks which is based on the fisher's score.		
	Correlation & Coefficient	Finds the variable which perform well and highly correlated with the target variable.		
Wrapper [28][37]	SFS(Sequential Search)	Starts from the empty set and add the variable permanently if its result is more accurate than the previous one.	It consider the feature dependencies as while interacting with the classifier and it is not better computationally intensive.	This method contain the risk of the over fitting and provide the better result than filter method but consumes more time than the filter method.
	SBS(Sequential Search)	Starts with full set and remove the variable which are less effective to the target variable.		
	Genetic algorithm (Heuristic Search)	It utilizes an approximation technique, in which a set of possible solutions evolves via natural selection		
Embedded [39]	Regularization	It reduces the computation time of the wrapper method.	Performs more computationally intensive operations and interact with the classifier.	Increasing the weight for models by over fitting. Selection of the features is classifier dependent.
Hybrid [40]	Using Filter & Wrapper	It has better computational time and it takes advantage of the filter and wrapper method.	Provide more accuracy and flexibility to interact with other method.	Complexity of the hybrid method is increased as compare to the filter and wrapper method.
	Using Embedded & Wrapper	It has an edge over embedded and wrapper method.		

IV. CONCLUSION

Nowadays, a huge amount of data is produced and the process of finding the important content or noise free data from this enormous quantity is a bit tedious task. To overcome this

problem we can apply feature selection on the dataset. Through feature selection one can extract the important features which are relevant to the objective instead of considering all the features which are not important and irrelevant. It helps to reduce the computational time, identify the irrelevant features

and finally generate well classified models. This paper is an effort to elaborate the importance of feature selection, methods, techniques and algorithms used in feature selection. Further, the comparative study of various feature selection methods has been presented along with advantages and research gaps.

REFERENCES

- [1] J. Cai, J. Luo, S. Wang, and S. Yang, "Neurocomputing Feature selection in machine learning: A new perspective," *Neurocomputing*, vol. 300, pp. 70–79, 2018, doi: 10.1016/j.neucom.2017.11.077.
- [2] K. P. Singh, N. Basant, and S. Gupta, "Support vector machines in water quality management," *Anal. Chim. Acta*, vol. 703, no. 2, pp. 152–162, 2011, doi: 10.1016/j.aca.2011.07.027.
- [3] A. L. Blum and P. Langley, "Artificial Intelligence Selection of relevant features and examples in machine," *Artif. Intell.*, vol. 97, no. 1–2, pp. 245–271, 1997.
- [4] J. Miao and L. Niu, "A Survey on Feature Selection," *Procedia - Procedia Comput. Sci.*, vol. 91, no. Itqm, pp. 919–926, 2016, doi: 10.1016/j.procs.2016.07.111.
- [5] B. Xue, M. Zhang, W. N. Browne, and X. Yao, "A Survey on Evolutionary Computation Approaches to Feature Selection," *IEEE Trans. Evol. Comput.*, vol. 20, no. 4, pp. 606–626, 2016, doi: 10.1109/TEVC.2015.2504420.
- [6] Z. Zhao, F. Morstatter, S. Sharma, S. Alelyani, A. Anand, and H. Liu, "Advancing Feature Selection Research," *ASU Featur. Sel. Repos. Arizona State Univ.*, pp. 1–28, 2010, [Online]. Available: http://featureselection.asu.edu/featureselection_techreport.pdf.
- [7] K. Naidu, A. Dhenge, and K. Wankhade, "Feature selection algorithm for improving the performance of classification: A survey," *Proc. - 2014 4th Int. Conf. Commun. Syst. Netw. Technol. CSNT 2014*, pp. 468–471, 2014, doi: 10.1109/CSNT.2014.99.
- [8] S. Tiwari, S. Kumar, and K. Guleria, "Outbreak Trends of Coronavirus Disease–2019 in India: A Prediction," *Disaster Medicine and Public Health Preparedness*, vol. 14, no. 5, pp. e33–e38, 2020.
- [9] K. Guleria, A. Sharma, U. K. Lilhore, D. Prasad, "Breast Cancer Prediction and Classification Using Supervised Learning Techniques," *Journal of Computational and Theoretical Nanoscience*, 2020, 17(6), 2519–2522.
- [10] U. K. Lilhore, S. Simaiya, D. Prasad, and K. Guleria, "A Hybrid Tumour Detection and Classification Based on Machine Learning," *J. Comput. Theor. Nanosci.*, vol. 17, no. 6, pp. 2539–2544, 2020, doi: 10.1166/jctn.2020.8927.
- [11] N. K. Trivedi, S. Simaiya, U. K. Lilhore, and S. K. Sharma, "An efficient credit card fraud detection model based on machine learning methods," *Int. J. Adv. Sci. Technol.*, vol. 29, no. 5, pp. 3414–3424, 2020.
- [12] S. K. Sharma, U. K. Lilhore, S. Simaiya, and N. K. Trivedi, "An Improved Random Forest Algorithm for Predicting the COVID -19 Pandemic Patient Health," vol. 25, no. 1, pp. 67–75, 2021.
- [13] A. Anand, N.K. Trivedi, A. Jha, V. Singh, "A Sentiment Analysis-Based Approach to Predict Cryptocurrency Ethereum Price using Machine Learning," *Test Engineering and Management*, Vol. 83: March/April 2020.
- [14] L. Song, A. Smola, A. Gretton, K. M. Borgwardt, and J. Bedo, "Supervised feature selection via dependence estimation," *ACM Int. Conf. Proceeding Ser.*, vol. 227, pp. 823–830, 2007, doi: 10.1145/1273496.1273600.
- [15] J. C. Ang, A. Mirzal, H. Haron, and H. N. A. Hamed, "Supervised, unsupervised, and semi-supervised feature selection: A review on gene selection," *IEEE/ACM Trans. Comput. Biol. Bioinforma.*, vol. 13, no. 5, pp. 971–989, 2016, doi: 10.1109/TCBB.2015.2478454.
- [16] P. Mitra, S. Member, C. A. Murthy, and S. K. Pal, "Using Feature Similarity," vol. 24, no. 3, pp. 301–312, 2002.
- [17] N. Sánchez-Marroño, A. Alonso-Betanzos, and M. Tombilla-Sanromán, "Filter methods for feature selection - A comparative study," *Lect. Notes Comput. Sci. (including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics)*, vol. 4881 LNCS, pp. 178–187, 2007, doi: 10.1007/978-3-540-77226-2_19.
- [18] H. A. Le Thi, V. V. Nguyen, and S. Ouchani, "Gene selection for cancer classification using DCA," *Lect. Notes Comput. Sci. (including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics)*, vol. 5139 LNAI, pp. 62–72, 2008, doi: 10.1007/978-3-540-88192-6-8.
- [19] R. Kohavi and G. H. John, "Wrappers for feature subset selection," *Artif. Intell.*, vol. 97, no. 1–2, pp. 273–324, 1997, doi: 10.1016/s0004-3702(97)00043-x.
- [20] S. M. Anouncia and U. K. Wiil, *Knowledge computing and its applications: Knowledge computing in specific domains*, vol. 2. Springer Singapore, 2018.
- [21] T. M. Khoshgoftaar, A. Fazelpour, H. Wang, and R. Wald, "A survey of stability analysis of feature subset selection techniques," *Proc. 2013 IEEE 14th Int. Conf. Inf. Reuse Integr. IEEE IRI 2013*, pp. 424–431, 2013, doi: 10.1109/IRI.2013.6642502.
- [22] R. K. Bania, "Survey on Feature Selection for Data Reduction," *Int. J. Comput. Appl.*, vol. 94, no. 18, pp. 1–7, 2014, doi: 10.5120/16456-2390.
- [23] D. R. Liu, H. L. Li, and D. Wang, "Feature selection and feature learning for high-dimensional batch reinforcement learning: A survey," *Int. J. Autom. Comput.*, vol. 12, no. 3, pp. 229–242, 2015, doi: 10.1007/s11633-015-0893-y.
- [24] A. H. Shahana and V. Preeja, "Survey on Feature Subset Selection for," *Int. Conf. Circuit, Power Comput. Technol.*, p. 4, 2016.
- [25] W. C. Mlambo, N., "A survey and comparative study of filter and wrapper feature selection techniques," *Int. J. Eng. Sci.*, vol. 5, no. 8, pp. 57–67, 2016, [Online]. Available: <https://pdfs.semanticscholar.org/fe2c/a244dd5473270ce0a8e2a387349b9831cf47.pdf>.
- [26] J. Suto, S. Oniga, and P. P. Sitar, "Comparison of wrapper and filter feature selection algorithms on human activity recognition," *2016 6th Int. Conf. Comput. Commun. Control. ICCCC 2016*, no. Iccccc, pp. 124–129, 2016, doi: 10.1109/ICCC.2016.7496749.
- [27] S. García, J. Luengo, F. Herrera, S. García, J. Luengo, and F. Herrera,

“Feature selection,” *Intell. Syst. Ref. Libr.*, vol. 72, no. 6, pp. 163–193, 2015, doi: 10.1007/978-3-319-10247-4_7.

- [28] A. S. Abdullah, C. Ramya, V. Priyadharsini, C. Reshma, and S. Selvakumar, “A survey on evolutionary techniques for feature selection,” *2017 Conf. Emerg. Devices Smart Syst. ICEDSS 2017*, no. March, pp. 58–62, 2017, doi: 10.1109/ICEDSS.2017.8073659.
- [29] D. Panday, R. Cordeiro de Amorim, and P. Lane, “Feature weighting as a tool for unsupervised feature selection,” *Inf. Process. Lett.*, vol. 129, pp. 44–52, 2018, doi: 10.1016/j.ipl.2017.09.005.
- [30] N. Krishnaveni, “Feature Selection Algorithms for Data Mining Classification: A Survey,” *Indian J. Sci. Technol.*, vol. 12, no. 1, pp. 1–11, 2019, doi: 10.17485/ijst/2018/v12i6/139581.
- [31] V. Kumar, “Feature Selection: A literature Review,” *Smart Comput. Rev.*, vol. 4, no. 3, 2014, doi: 10.6029/smarter.2014.03.007.
- [32] G. Chandrashekar and F. Sahin, “A survey on feature selection methods,” *Comput. Electr. Eng.*, vol. 40, no. 1, pp. 16–28, 2014, doi: 10.1016/j.compeleceng.2013.11.024.
- [33] S. Noelia, “Filter Methods for Feature Selection – A,” pp. 178–187, 2007.
- [34] B. Azhagusundari and A. S. Thanamani, “Feature Selection based on Information Gain,” no. 2, pp. 18–21, 2013.
- [35] S. Bahassine, A. Madani, M. Al-sarem, and M. Kissi, “Feature selection using an improved Chi-square for Arabic text classification,” *J. King Saud Univ. - Comput. Inf. Sci.*, vol. 32, no. 2, pp. 225–231, 2020, doi: 10.1016/j.jksuci.2018.05.010.
- [36] R. Saidi, W. Bouaguel, and N. Essoussi, *on the Genetic Algorithm and Pearson Correlation Coefficient*. Springer International Publishing.
- [37] S. Maldonado and R. Weber, “A wrapper method for feature selection using Support Vector Machines,” *Inf. Sci. (Ny)*, vol. 179, no. 13, pp. 2208–2217, 2009, doi: 10.1016/j.ins.2009.02.014.
- [38] O. H. Babatunde, L. Armstrong, D. Diepeveen, B. Oluleye, A. Leisa, and D. Dean, “A Genetic Algorithm-Based Feature Selection A Genetic Algorithm-Based Feature Selection,” vol. 5, pp. 899–905, 2014.
- [39] T. N. Lal, O. Chapelle, and J. Weston, “Chapter 5 Embedded Methods,” vol. 165, pp. 137–138, 2006.
- [40] Y. Wang and L. Feng, “PT US CR,” *Expert Syst. Appl.*, 2018, doi: 10.1016/j.eswa.2018.01.041.